

Initial O+O EMCAL Calibration

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We use Pho3 and Pho4 for Run3 OO calibration

135 different runs from Run 82388 to 82703

π^0 target mass was calculated to be 0.140949264

rc->set_StringFlag("CDB_GLOBALTAG", "newcdbtag");

pt1, pt2 = 0.9, ΔR = 1.1, α cut = 0.6, cluster χ^2 cut = 0.05

π^0 Based EMCAL Calibration Strategy

- **First Step: Evaluate Target Mass**

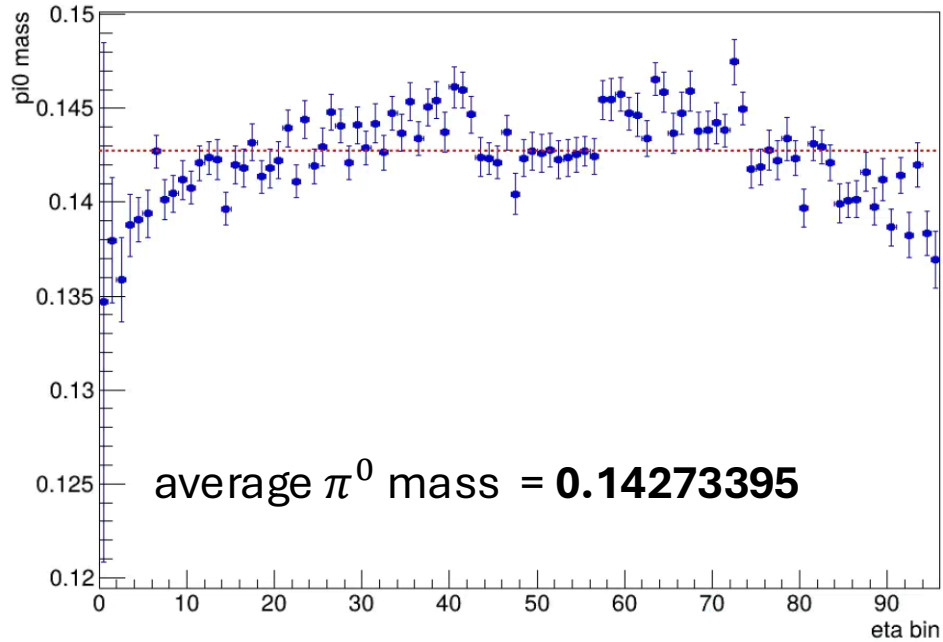
1. Use MB Trig. Data in run3 O-O to get the average (averaged over all EMCAL towers) π^0 mass
2. Use Pho3 + Pho4 Trig. Data in run3 O-O to get the average (averaged over all EMCAL towers) π^0 mass
3. Evaluate average (averaged over all EMCAL towers) π^0 mass in Run35 O-O sim.
4. Take the ratio of π^0 mass of MB to Pho3+Pho4 trig. Data and propagate it to the Run35 sim. To get the target mass to be used in Run3 O-O Calibration later.

- **Second Step: Iterative Calibration**

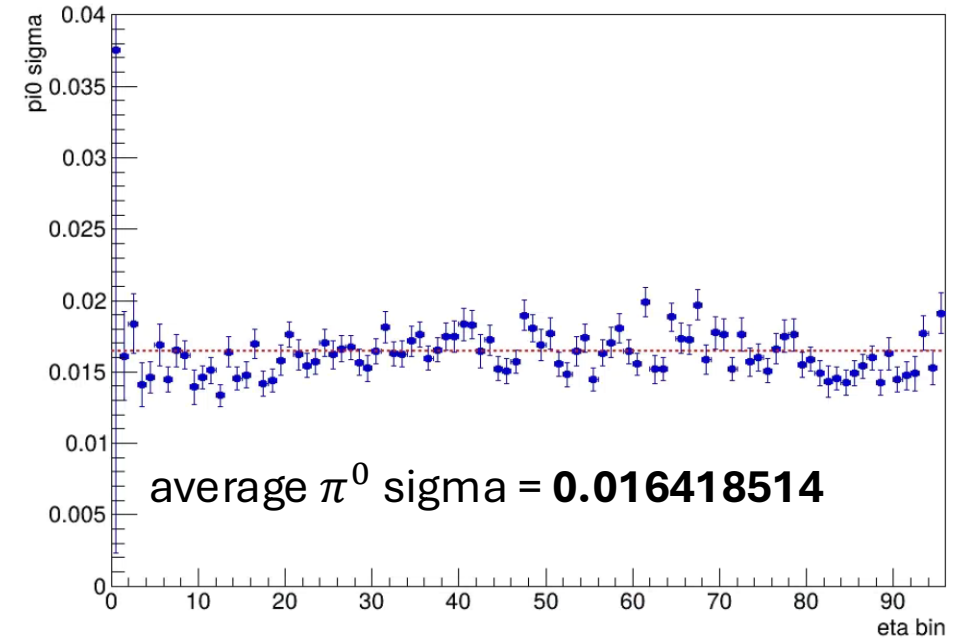
1. Use Pho3+Pho4 Trigger Data in Run3 O-O
2. Start with initial calibration
3. Run approximately 10 iterations
4. Update calibration constants (use the target mass calculated in the first step) after each iteration and use the updated calibration in subsequent iterations

π^0 mass and sigma Distribution in Run35 O-O Sim.

Peak vs eta bin, Run35 Sims



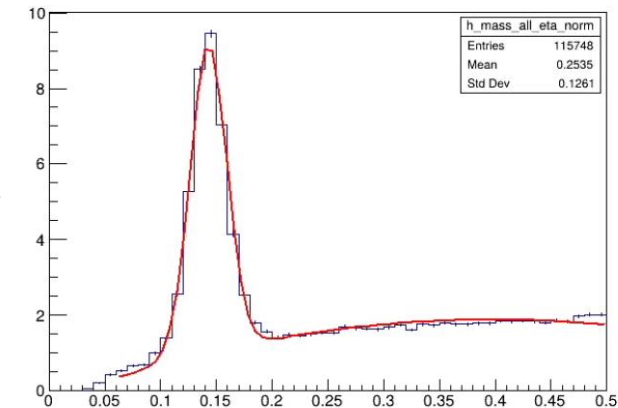
Sigma vs eta bin, Run35 Sims



Invariant Mass Hist for all EMCal towers combined and associated fit function

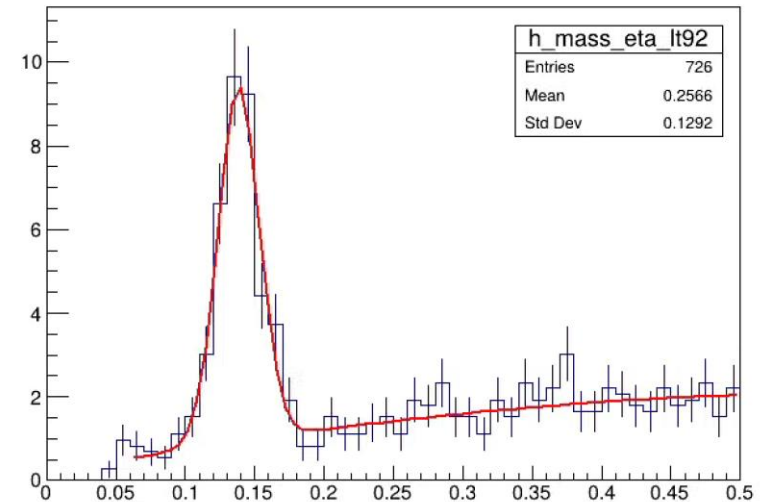
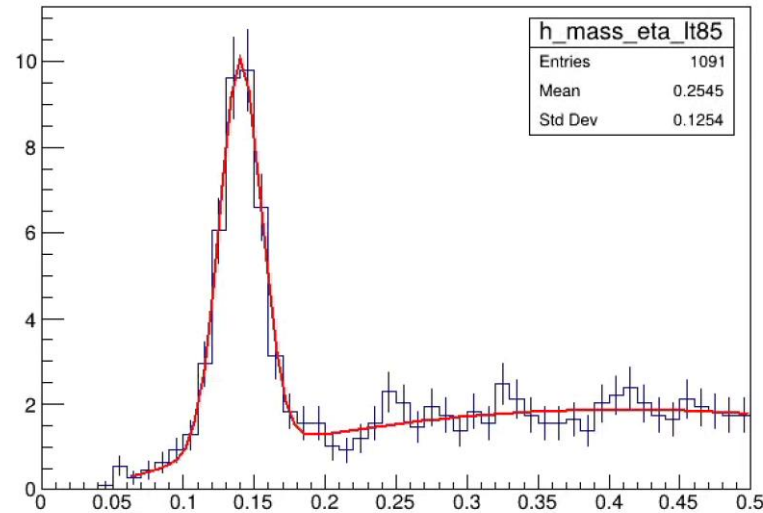
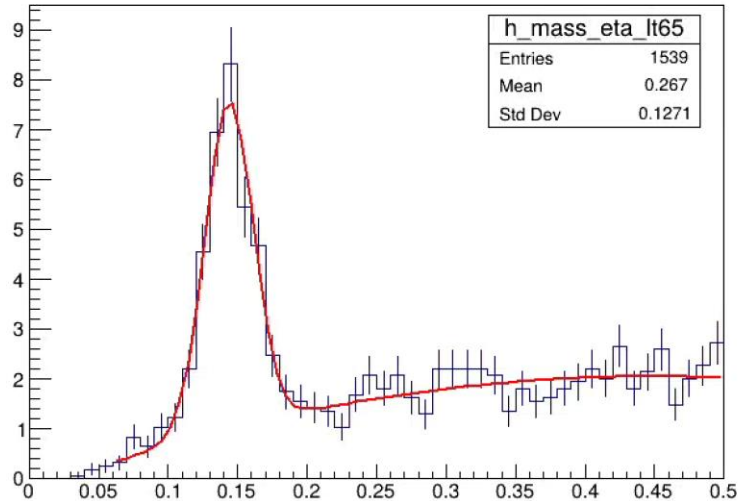
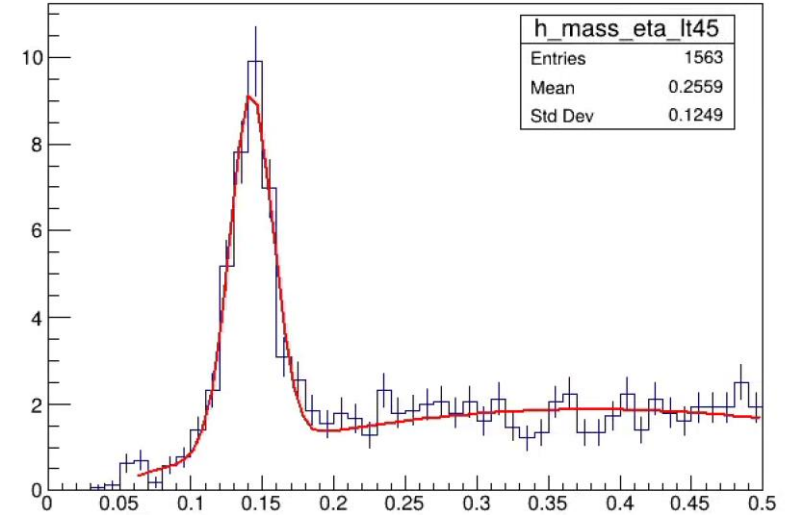
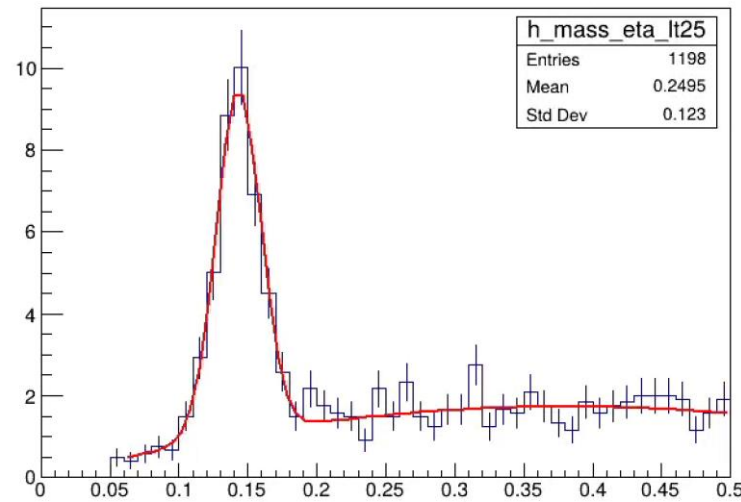
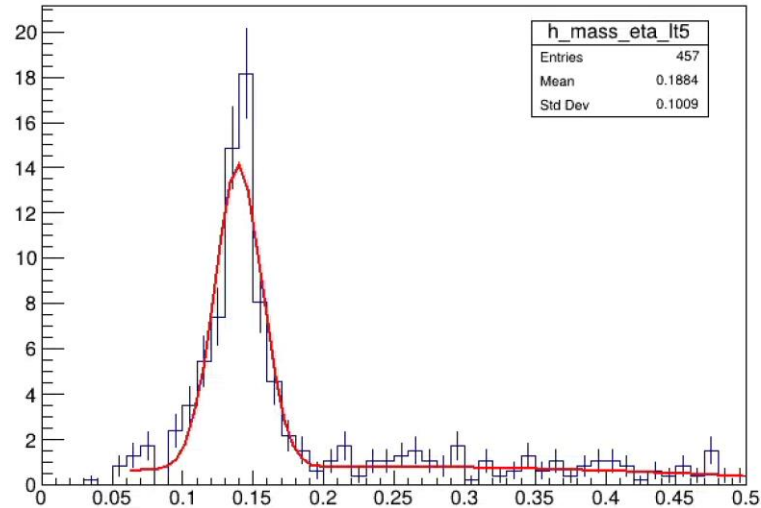


We use this fit to find the average π^0 mass and sigma (red horizontal line in both the above plots)

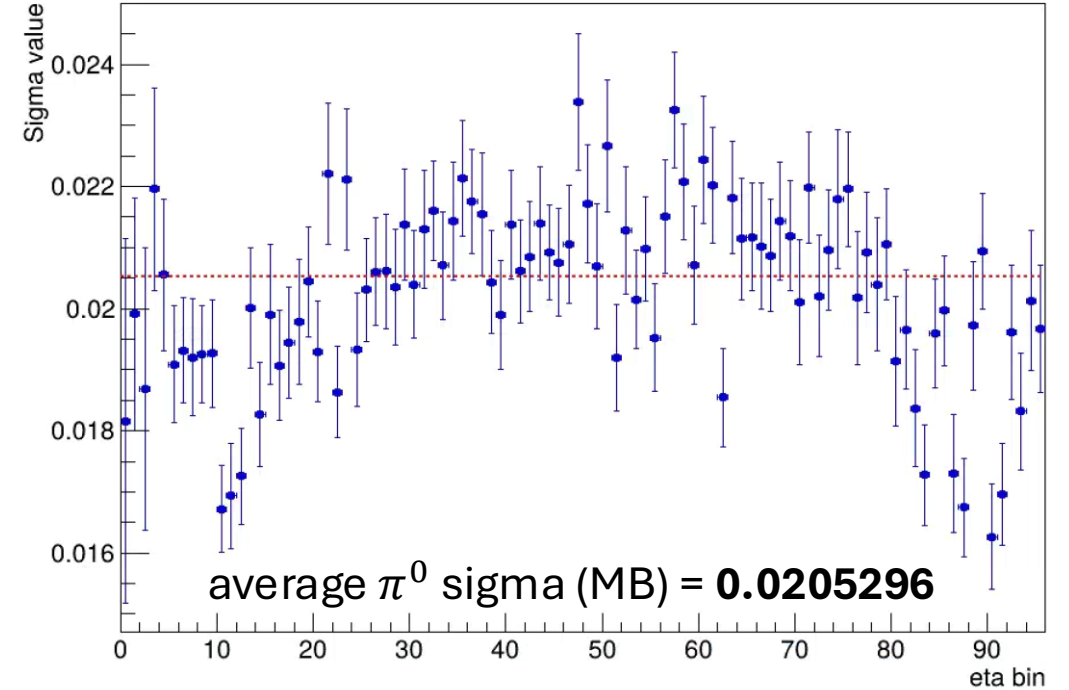
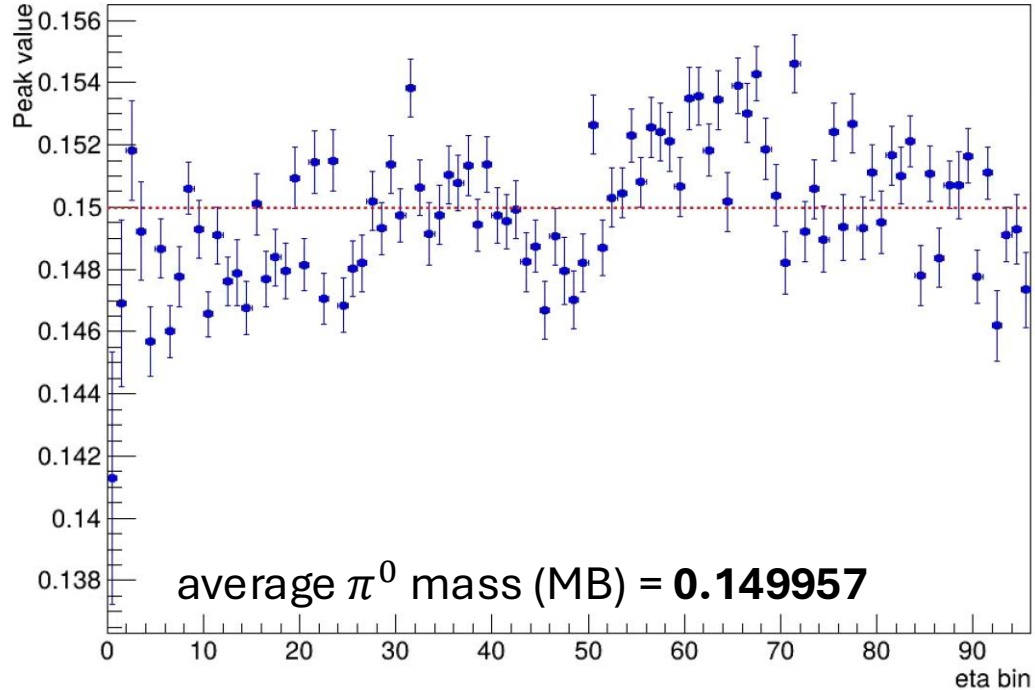


Some Fitting Examples in Run35 O-O sims

All of them are eta slice invariant mass histograms



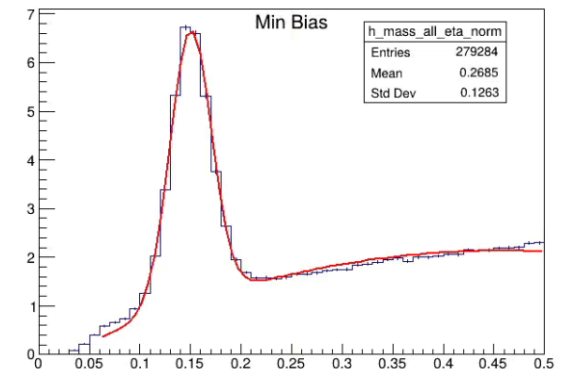
π^0 mass and sigma for Run3 O-O using MB



Invariant Mass Hist for all EMCAL towers combined and associated fit function

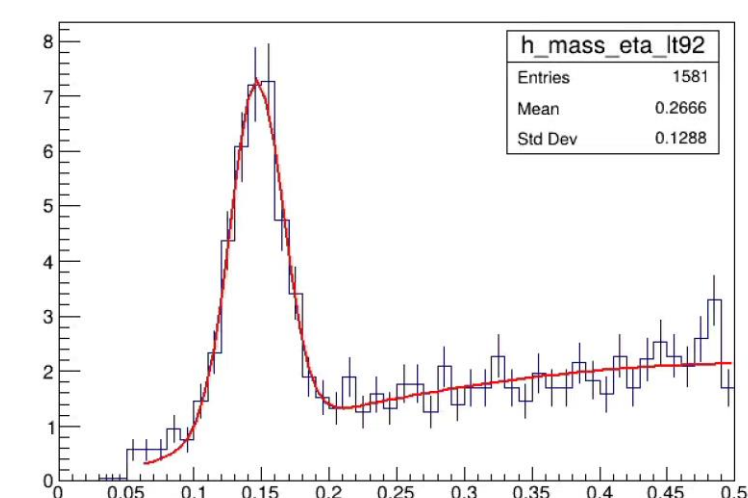
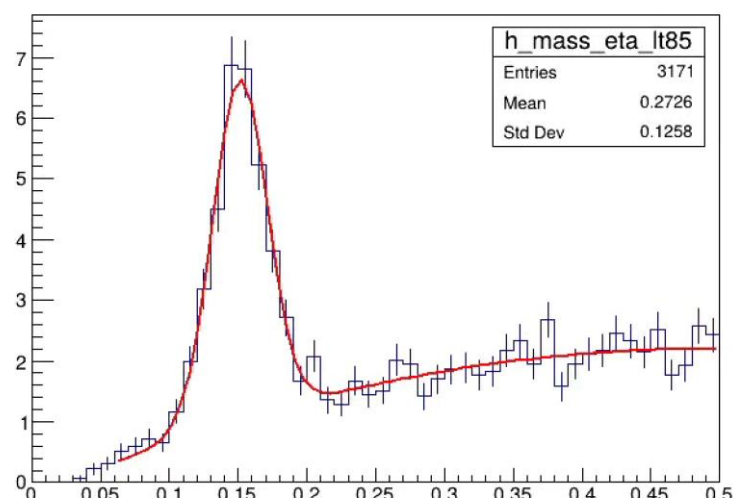
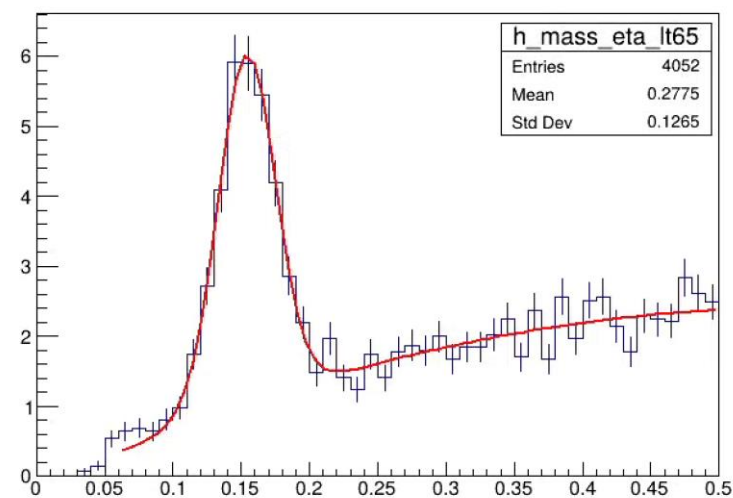
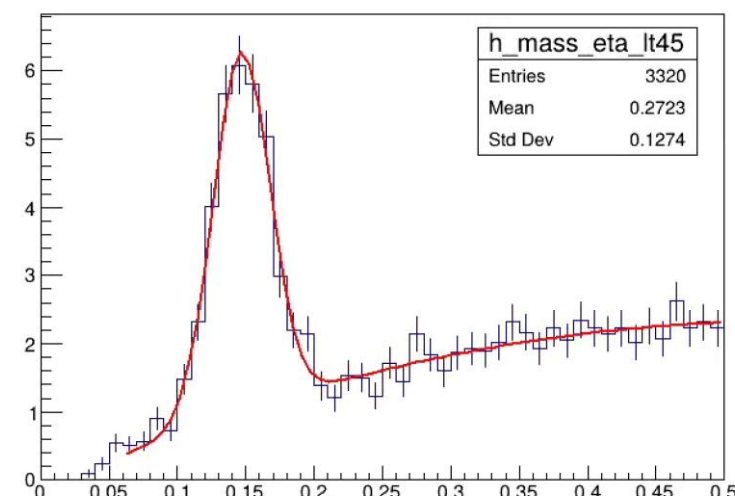
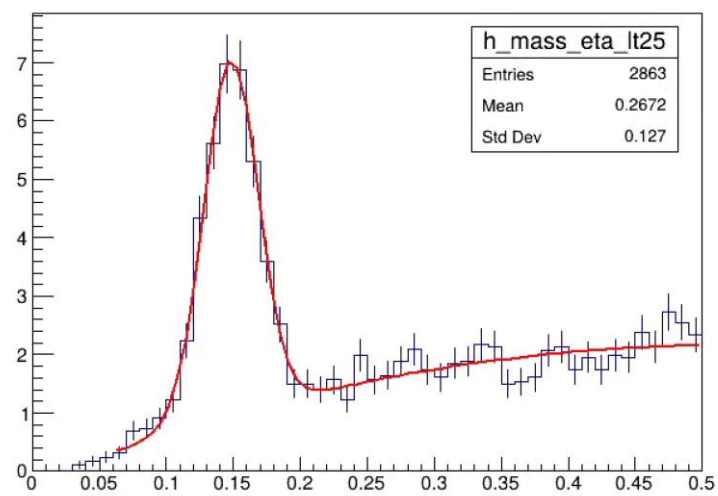
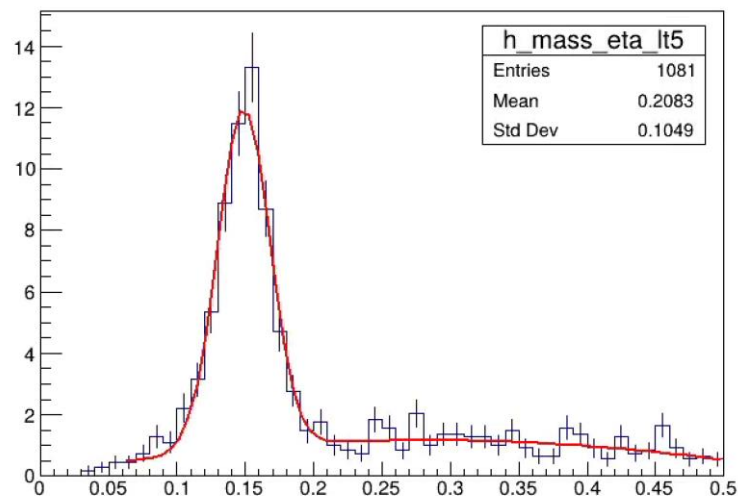


We use this fit to find the average π^0 mass and sigma (red horizontal line in both the above plots)

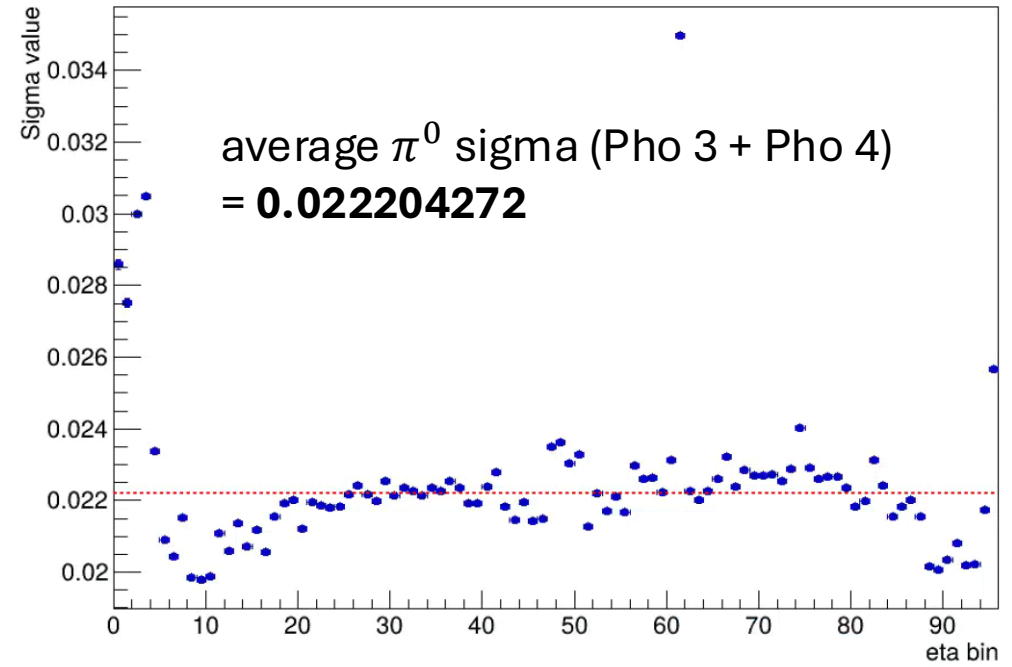
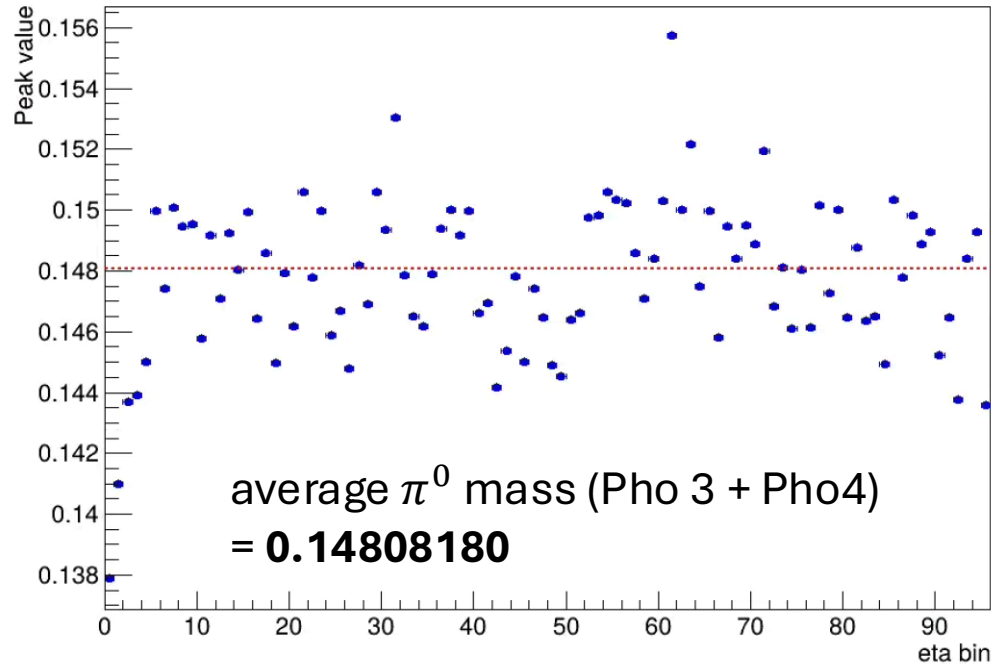


Some Fitting Examples in Run3 OO MB

Each of them are invariant mass histogram for a single eta-slice



π^0 mass and sigma for Run3 O-O using Pho3 + Pho4

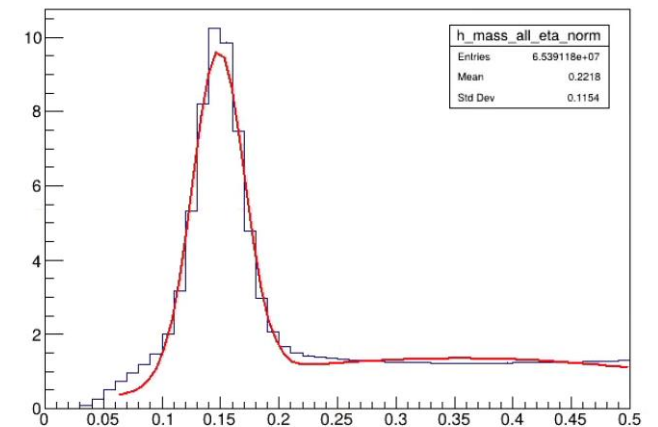


Invariant Mass Hist for all towers combined and associated fit function



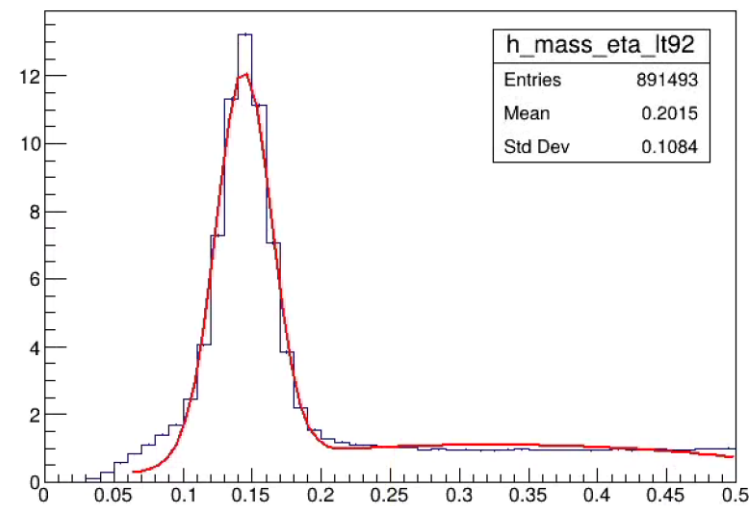
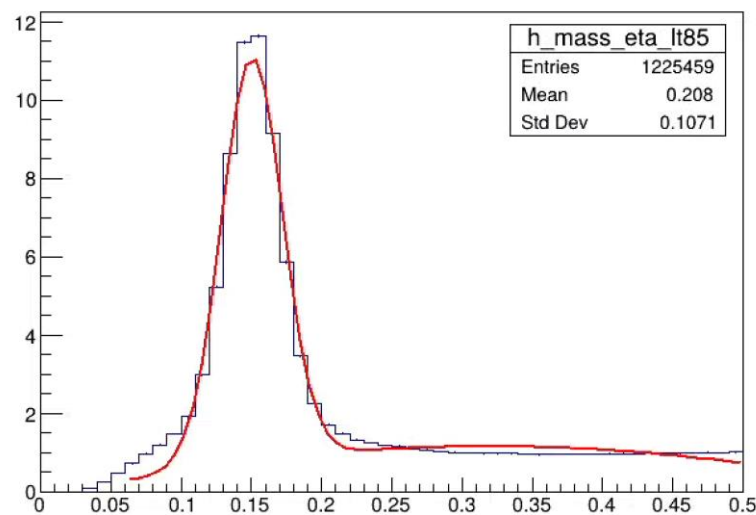
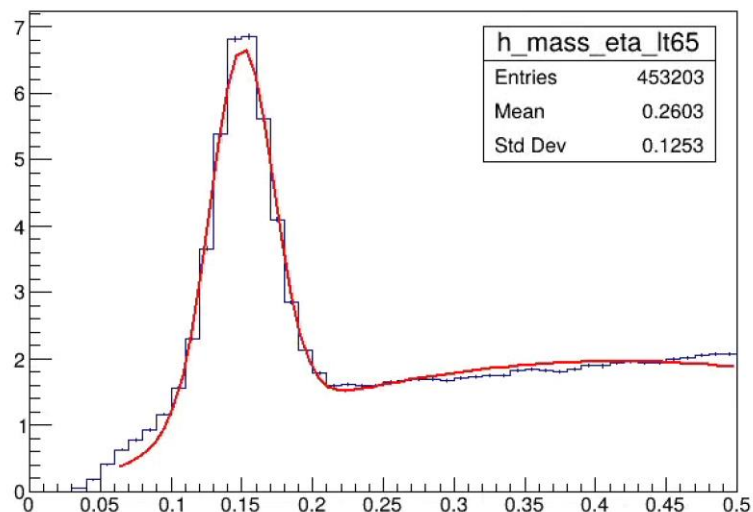
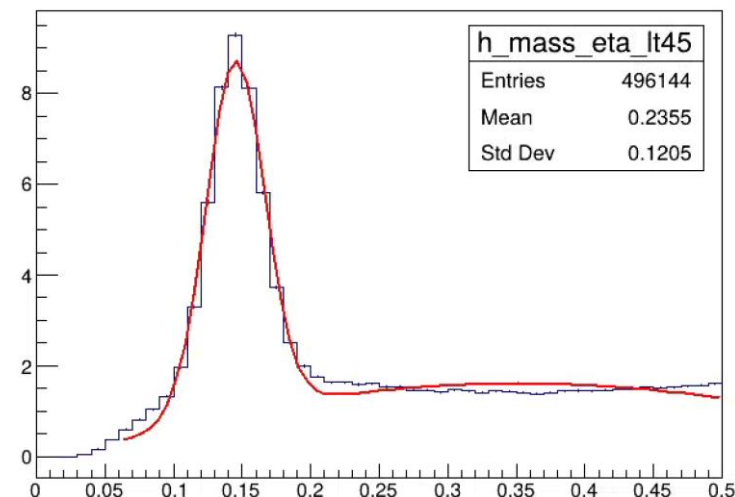
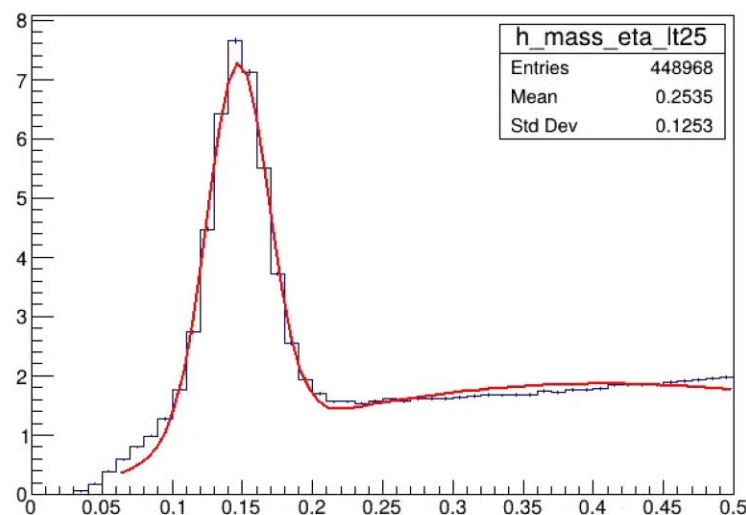
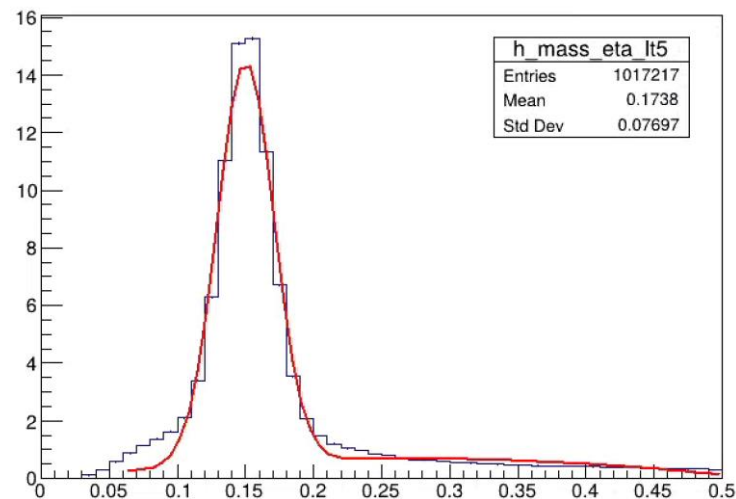
We use this to find the average π^0 mass and sigma (red horizontal line in both the above plots)

For low ieta bins (ieta = 0 – 3), a left-side tail in the π^0 peak is not well described by the current fit function, biasing the extracted mass low and the σ high (slide 9 for details).



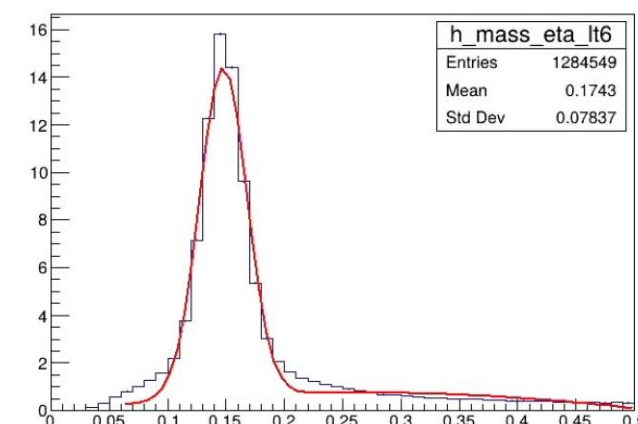
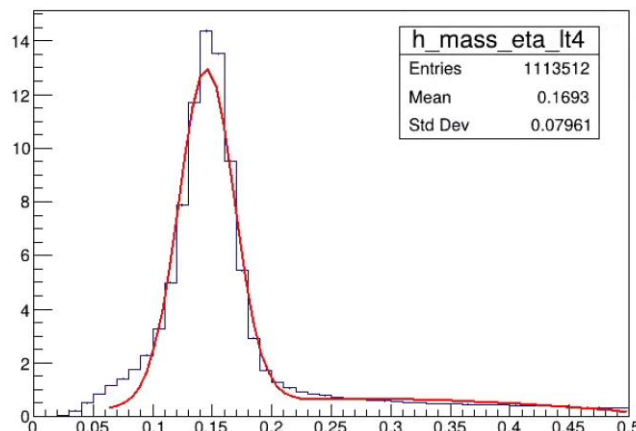
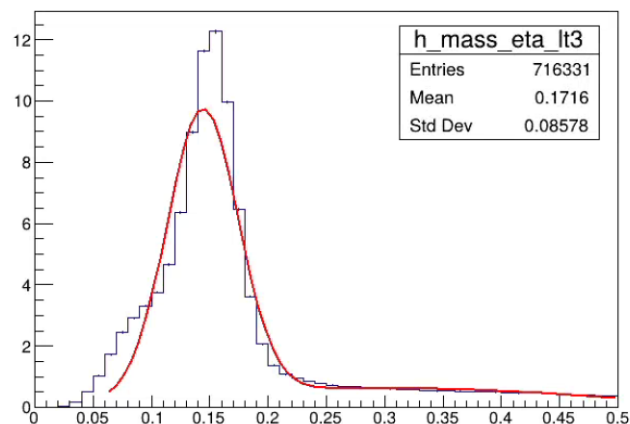
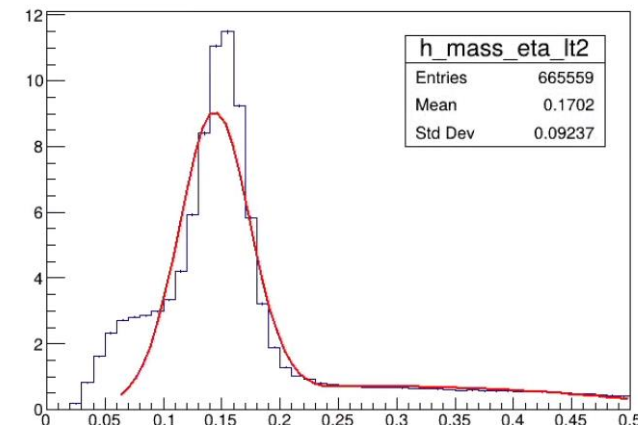
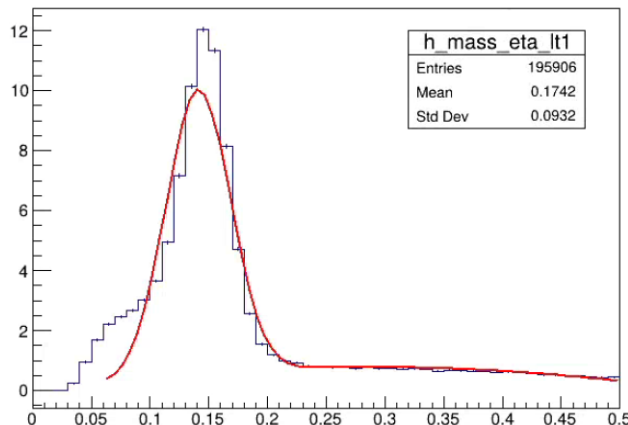
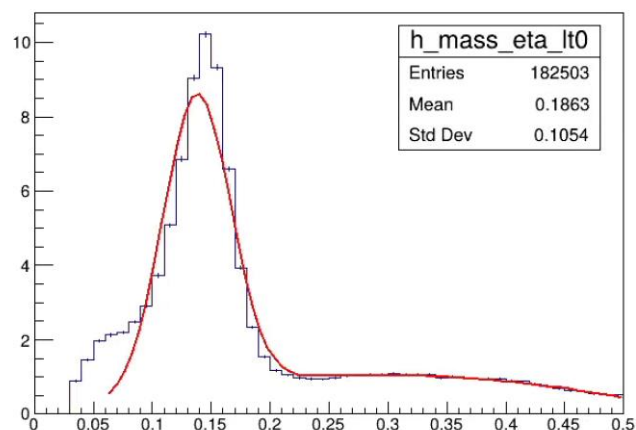
Some Fitting Examples in Run3 OO Pho3, 4

Each of them are invariant mass histogram for a single eta-slice



Some Fitting Examples in Run3 OO Pho3, 4

Each of them are invariant mass histogram for a single eta-slice for those few initial ieta bins

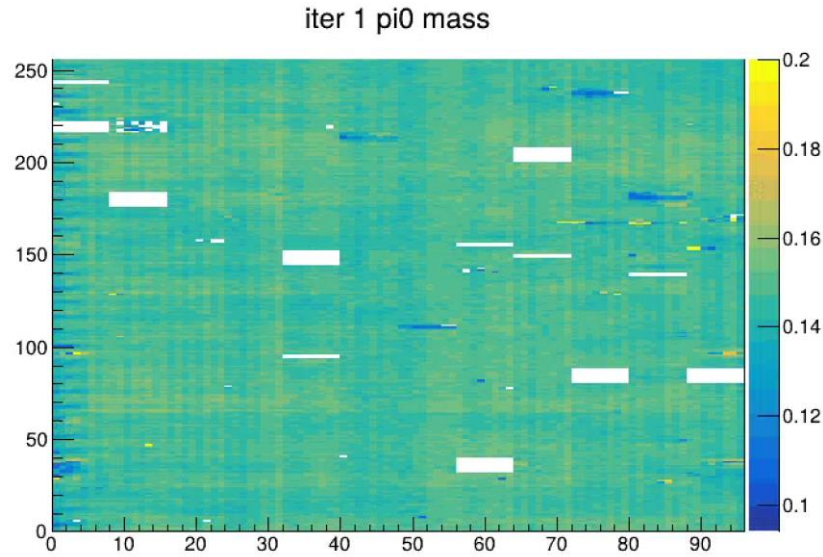


We observe a left-side tail in the π^0 peak, which is not well described by the current fit. As a result, the extracted sigma is higher, and the mean (mass) is slightly shifted to lower values. This effect is most pronounced in the first few eta bins and gradually diminishes at higher eta bins.

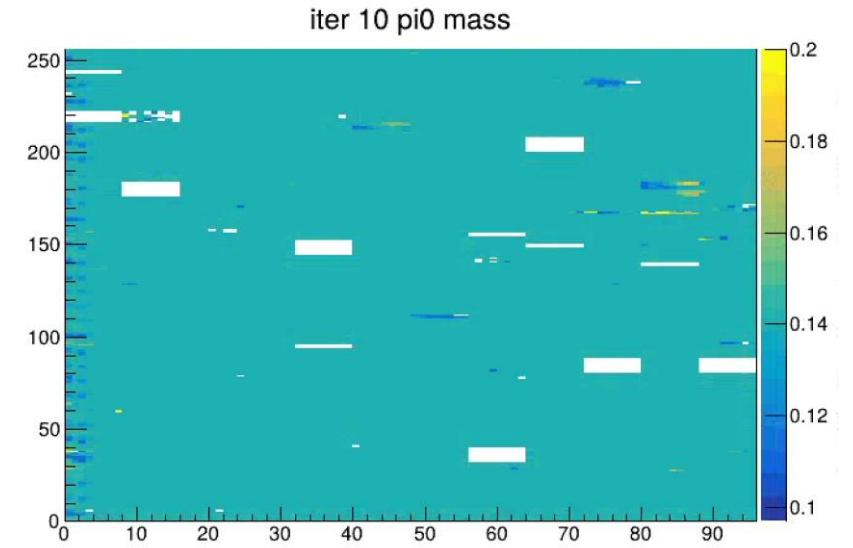
Evaluate Target Mass For Run3 O-O

- Since we are planning to use the Pho3+Pho4 trigger for π^0 calibration, we must evaluate a new target (as discussed in slide 2).
- In run35 O-O simulations, we observe π^0 mass (peak value) is **0.14273395** and π^0 sigma is **0.016418514**
- In run 3 O-O data, using MB Trigger, we get the following values of π^0 mass and sigma :
mass = **0.149957**, sigma = **0.0205296**
- In run 3 O-O data, using Pho3 + Pho4 Trigger, we get the following π^0 mass and sigma :
mass = **0.148082** , sigma = **0.0222043**
- The ratio of π^0 mass for Pho3+Pho4 trigger event to MB trigger for O-O data is : **0.98749642**
- This set out the final target mass to be **0.140949264**

π^0 Mass and Sigma using Pho3+Pho4 Run3 O-O

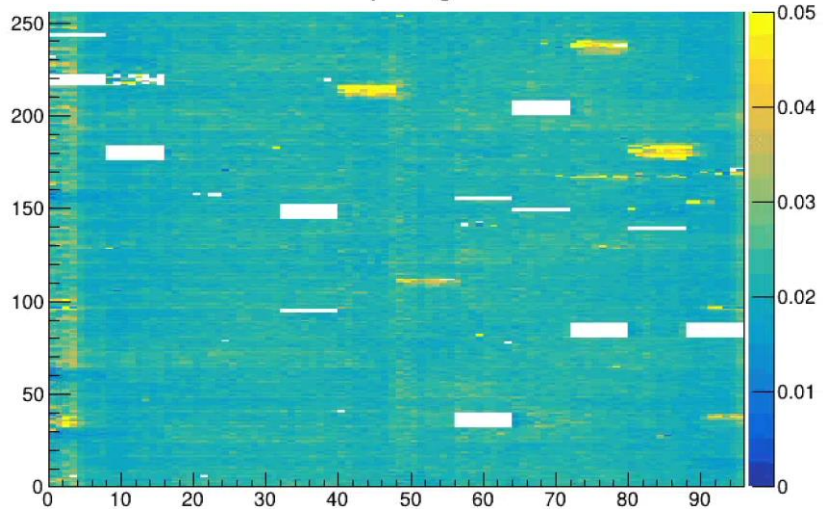


π^0 mass 1st iteration (left)
and 10th iteration (right)

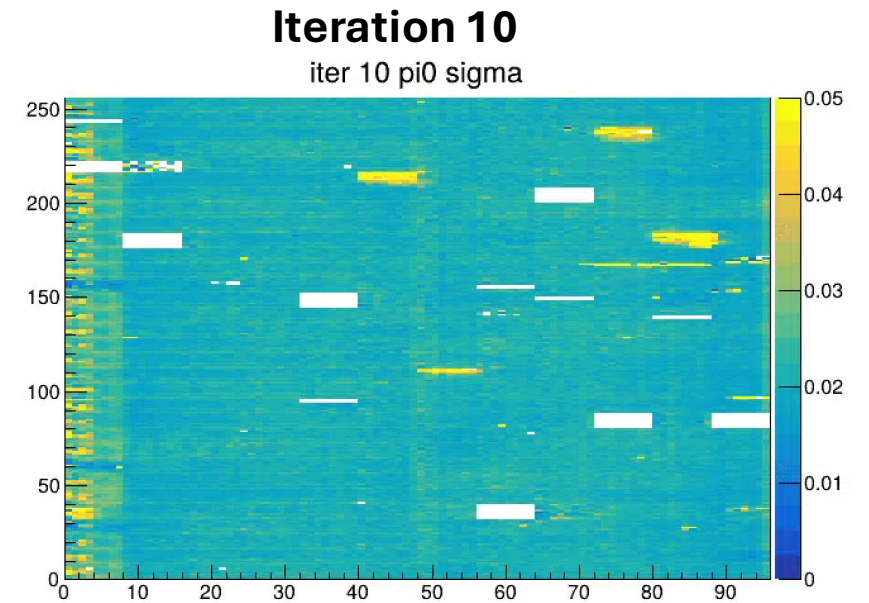


Iteration 1

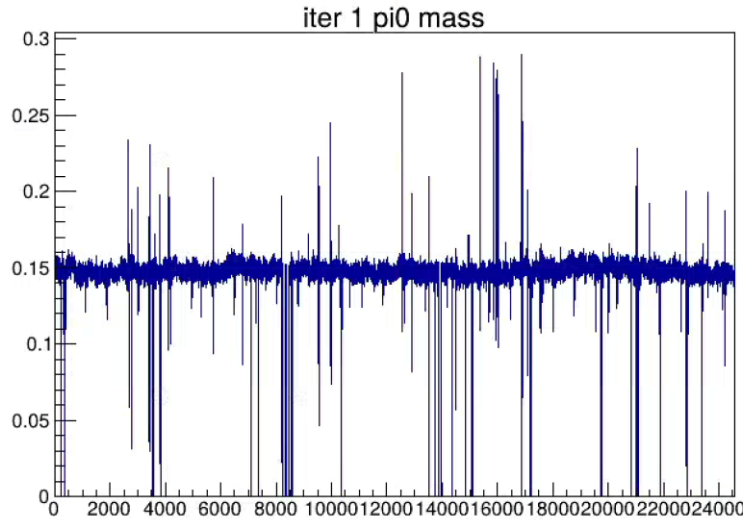
iter 1 pi0 sigma



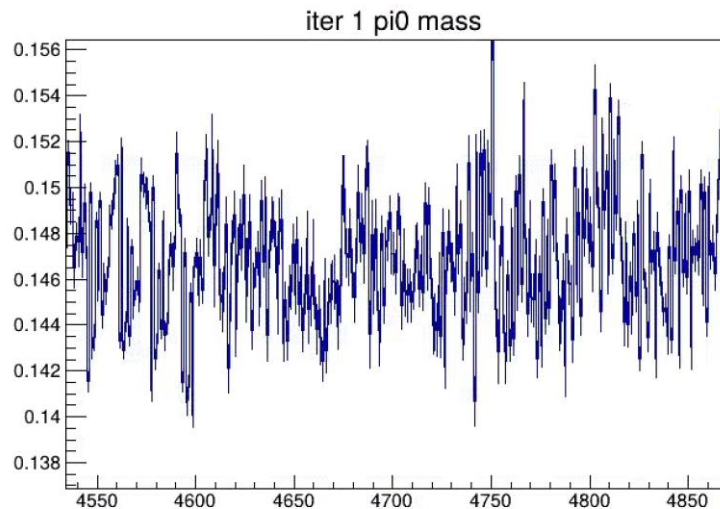
π^0 sigma 1st iteration (left)
and 10th iteration (right)



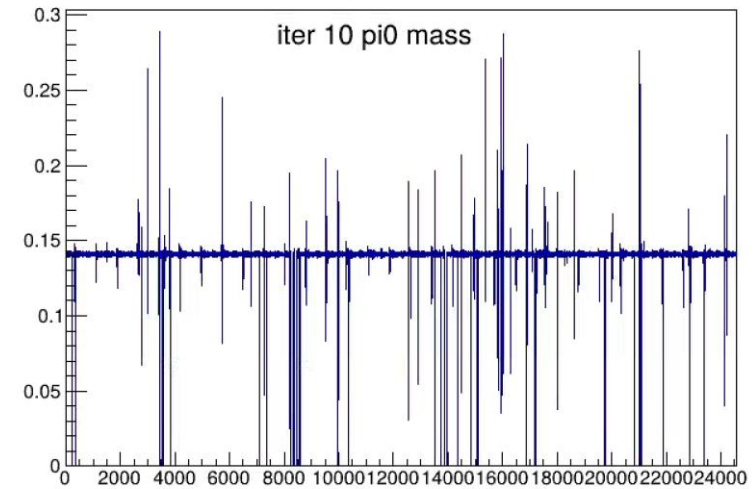
1D Hist of π^0 mass using Pho3+Pho4 Run3 O-O



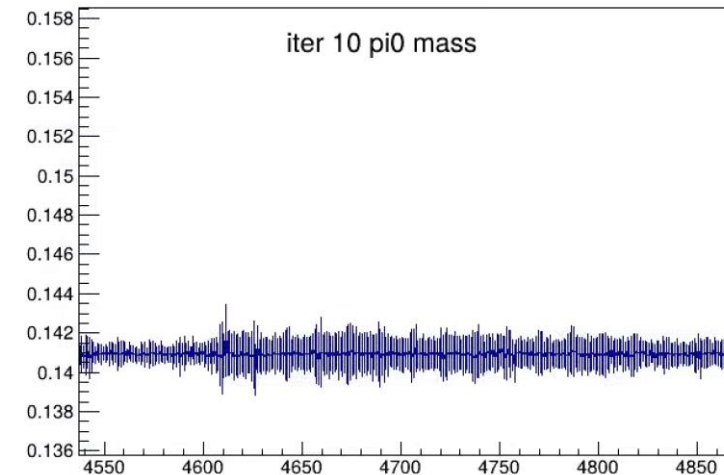
Below is zoomed in (along x-axis)
plot of 1D hist from above



1D Hist for π^0 mass for
iteration 1 (left) and
iteration 10 (right)



Below is zoomed in (along x-axis)
plot of 1D hist from above

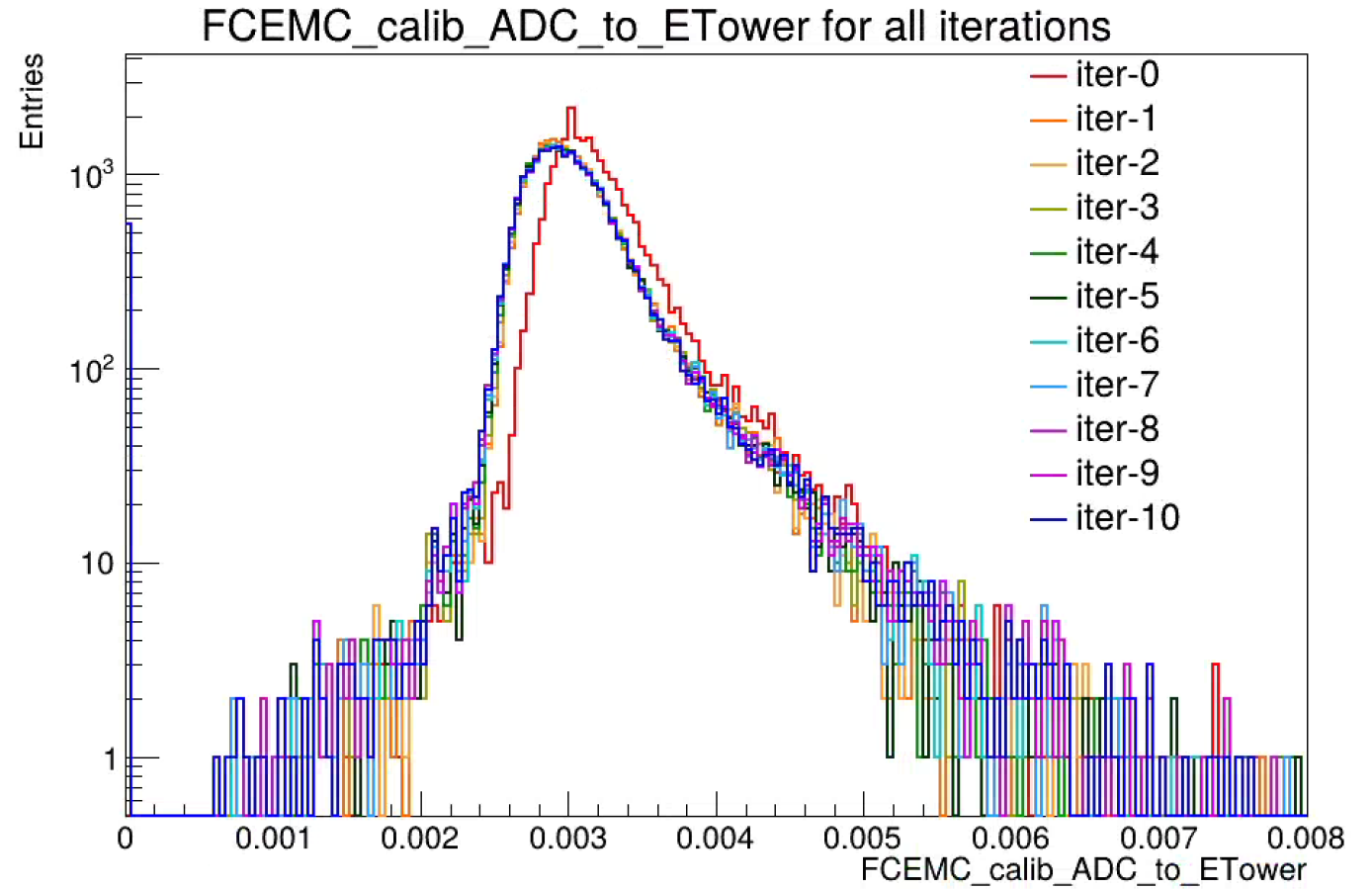


1D Hist for π^0 mass for
iteration 1 (left) and
iteration 10 (right)

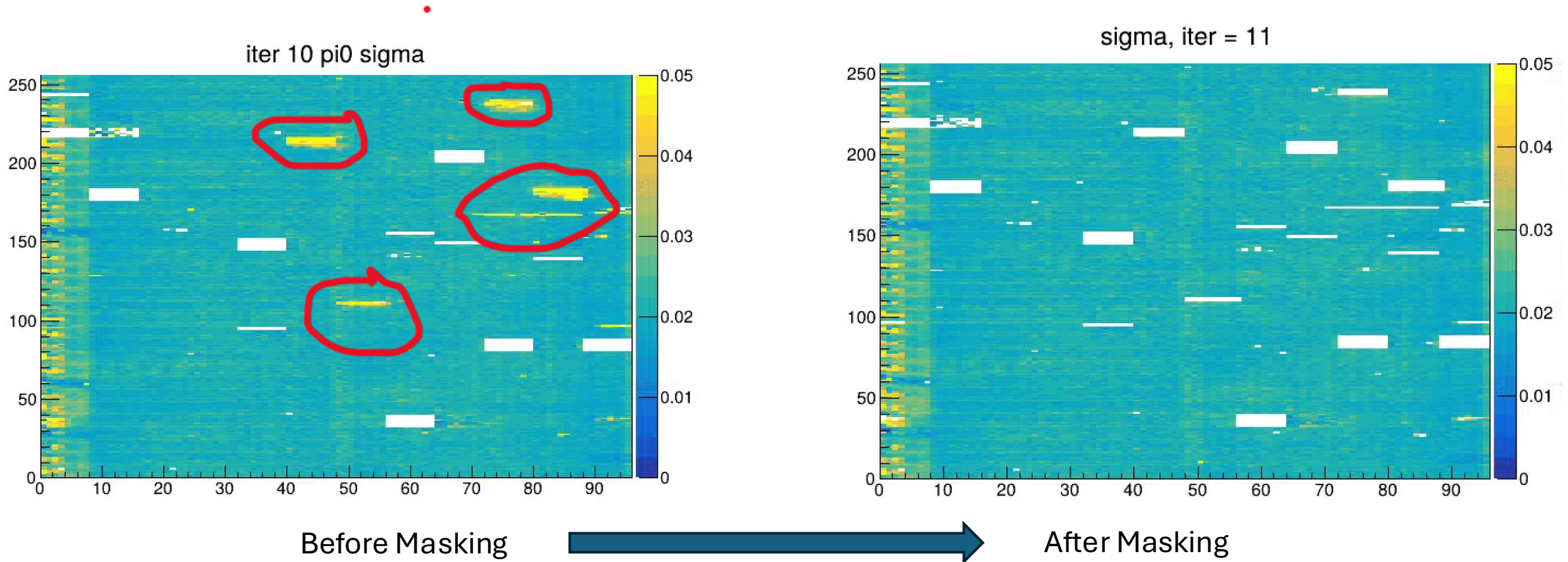


Convergence of Calibration Constants Across Iterations

- Iteration 0: Initial calibration
- Calibration constants evolve with iteration
- Distributions converge and stabilize by ~ iteration 10
- We didn't remove hard-to-calibrate region in this plot



Masking Hard-To-Calibrate Towers in Run3 O-O



From manual inspection, we have identified around 198 towers in 5 different regions as hard-to-calibrate towers (similar in run2pp)

As a temporary solution (could be made permanent ???) we masked those regions/area